

WEEK 3 (Day 5)

Matplotlib in Python

Matplotlib is a popular Python library used for data visualization. It helps create different types of graphs and charts such as bar charts, line graphs, pie charts, scatter plots, histograms, and box plots. It is widely used in Data Analytics, Data Science, and Machine Learning.

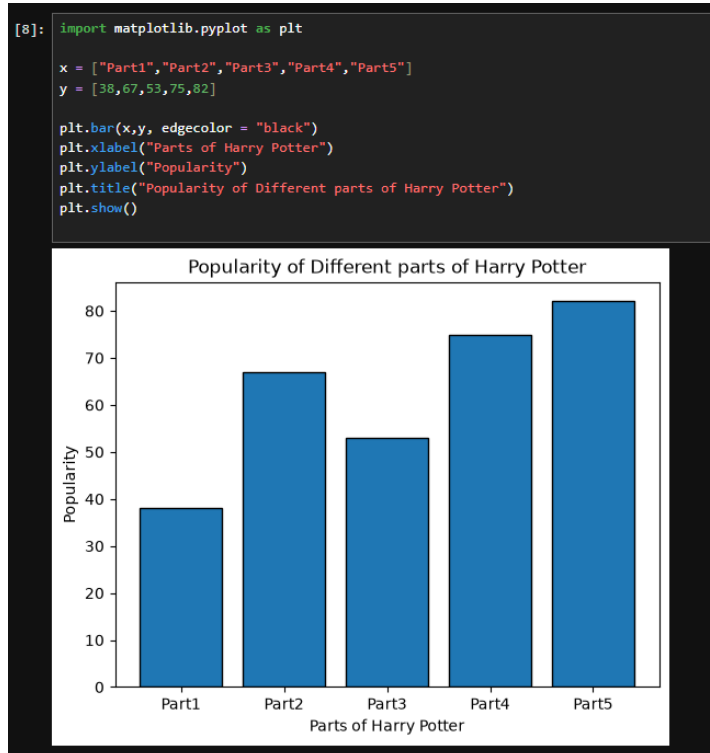
1. Line Plot

A **line plot** is used to show trends or changes in data over time.



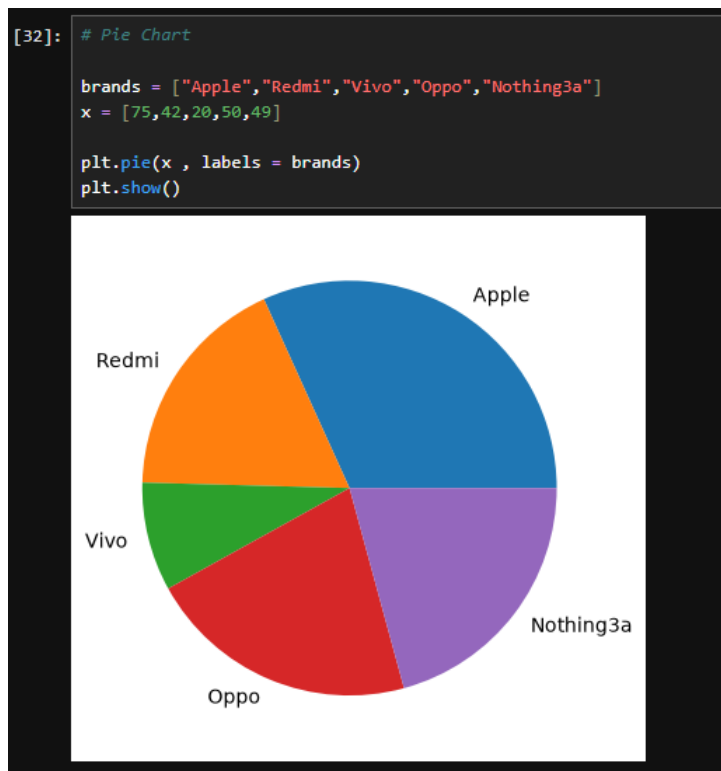
2. Bar Chart

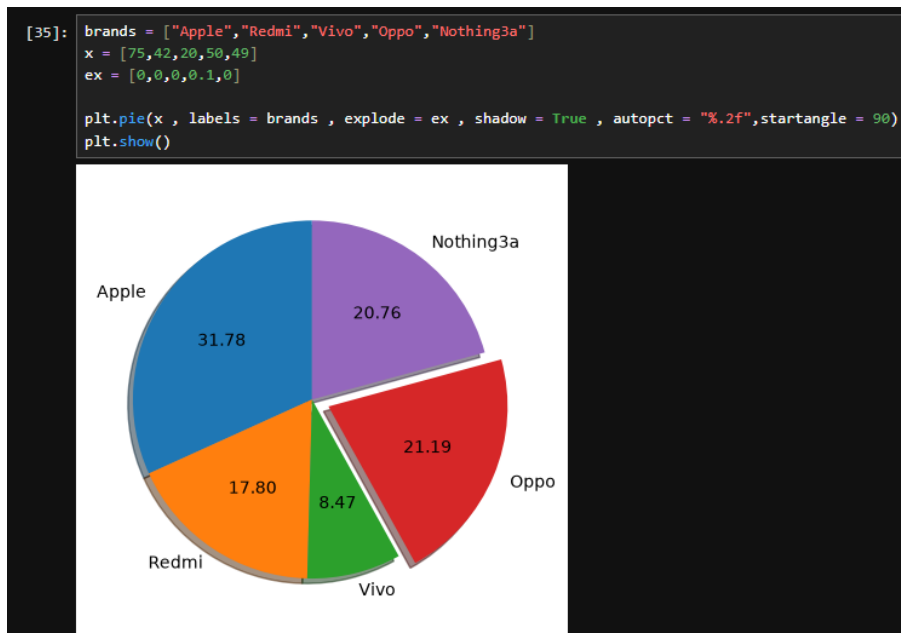
A **bar chart** is used to compare values among different categories.



3. Pie Chart

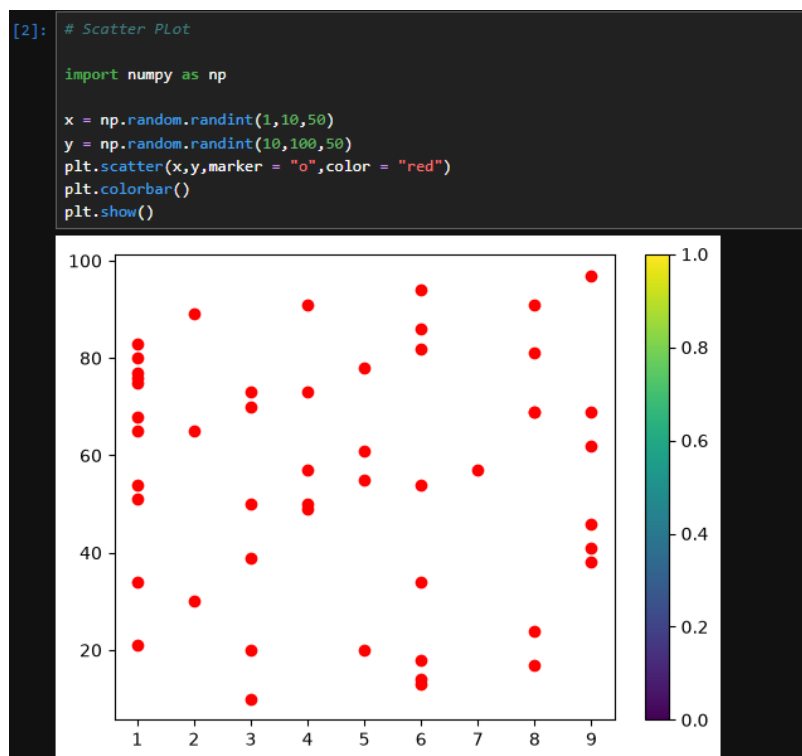
A **pie chart** shows the percentage or proportion of each category.





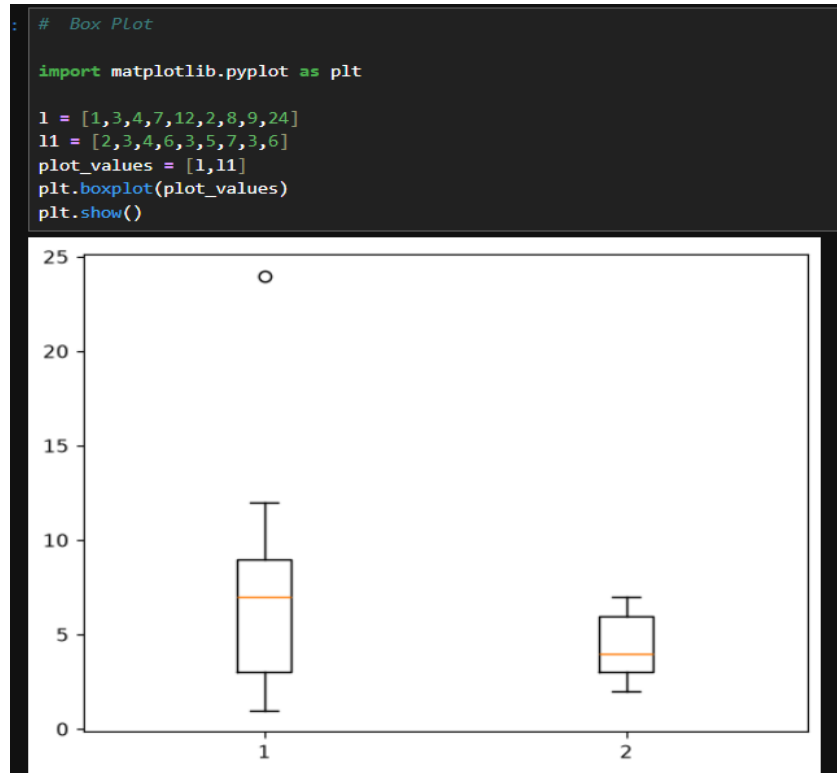
4. Scatter Plot

A **scatter plot** is used to show the relationship between two numerical variables.



5. Box Plot

A **box plot** is used to show the distribution of data and detect outliers.



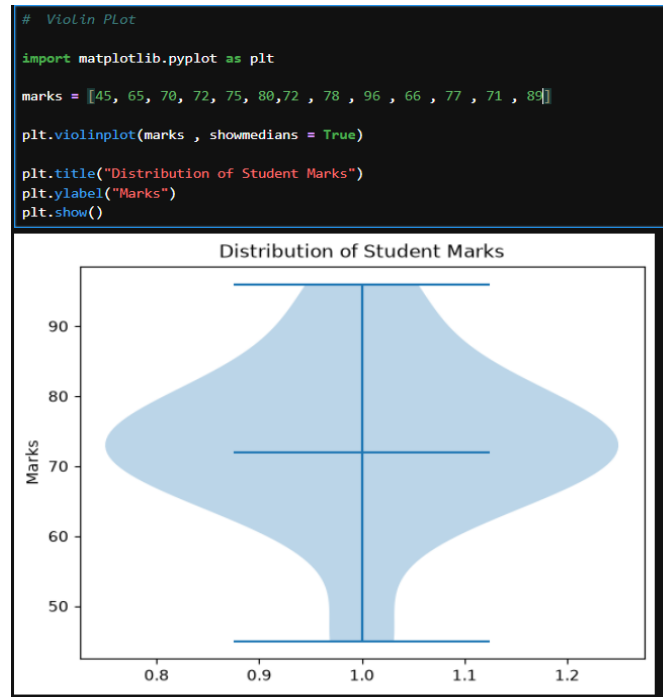
6. Histogram

A **histogram** is used to display the frequency distribution of continuous data.



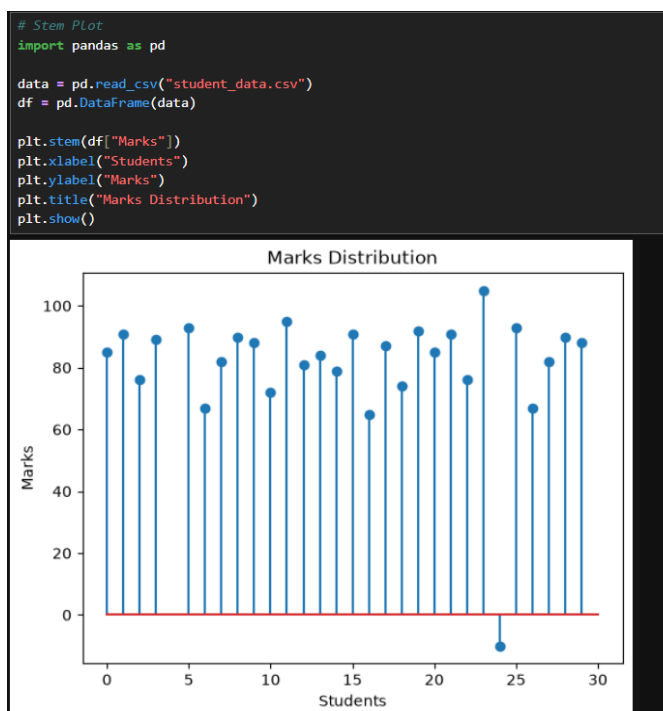
7. Violin Plot

A violin plot is a statistical graph used to visualize the distribution of numerical data. It combines the features of a box plot and a kernel density plot, showing both the summary statistics and the shape of the data distribution.



8. Stem Plot

A stem plot is a graph used to display discrete data. Each data point is represented by a vertical line (stem) extending from a baseline to the data value, with a marker at the top.



9. Stack Plot

A stack plot (also called a stacked area plot) is used to show how different components contribute to a total over time or across categories.

